



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

06-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS & TNAS] (SCORE-I)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	B,D	A,D	B,C	A,B,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	C	A	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	25.00	0.01	489.40 to 489.60	3.12 to 3.13	100.00	8.00	10.00	10.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B	A,B,D	A,B,C	A,C,D	B,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	D	C	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.20	189.00	2.00	3.00	367.50	0.25	305.00	85.71

PART-3 : MATHEMATICS

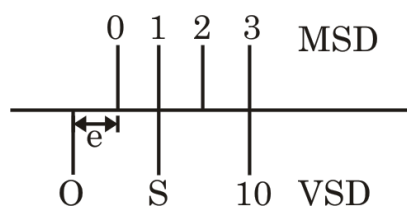
SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	B,C	A,C	A,C	A,B,C	A,B	B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	D	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	12.50	2.60	0.20	0.07	2.00	1275.00	7.00	0.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

2. Ans (B,D)

Given $4\text{MSD} = 10\text{VSD}$

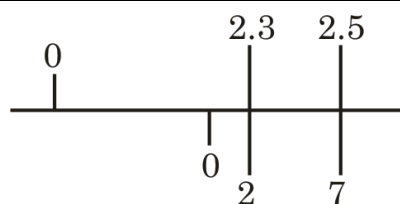
$$\Rightarrow 1\text{VSD} = 0.4\text{MSD}$$

$$\Rightarrow 1\text{VSD} = 0.4\text{mm}$$

$$\therefore 10\text{VSD} = e + 3\text{MSD}$$

$$\Rightarrow 10 \times 0.4 = e + 3$$

$$\Rightarrow e = 1\text{mm} \Rightarrow \text{-ve zero error}$$



$$25\text{mm} = \ell_m + 7\text{VSD}$$

$$\ell_m = 25 - 7 \times 0.4$$

$$\ell_m = 22.2\text{mm}$$

$$\therefore \text{Correct length} = \ell_m - \text{Zero error}$$

$$= \ell_m - (-1\text{mm})$$

$$\Rightarrow \ell = 23.2\text{mm}$$

3. Ans (A,D)

In hole is made in liquid of density ρ

$$R_1 = \sqrt{2gh} \sqrt{2 \times \frac{2h}{g}} = 2\sqrt{2}h$$

If hole is made in liquid of density 2ρ

$$R_2 = \sqrt{2 \times \left(\frac{5h}{2} - x\right) g \left(\frac{2x}{g}\right)}$$

$$R_2 = 2\sqrt{\left(\frac{5h}{2} - x\right) x}$$

$$(R_2)_{\max} = 2 \times \frac{5h}{4} = \frac{5h}{2}$$

$$x = \frac{5h}{4}$$

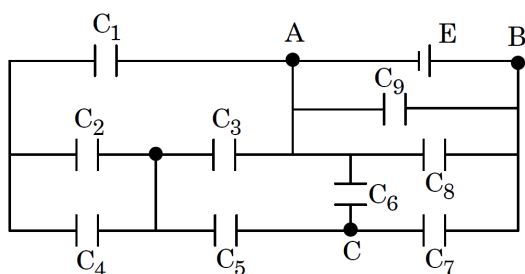
But if the hole is at interface then

$$R_{\max} = 2\sqrt{2}h > (R_2)_{\max}$$

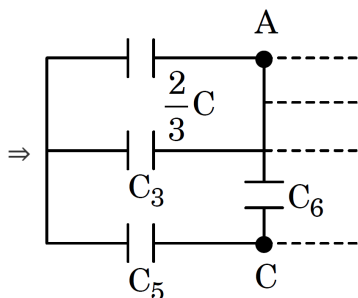
$$\therefore h_{\text{hole}} = 2h = 1\text{m}$$

4. Ans (B,C)

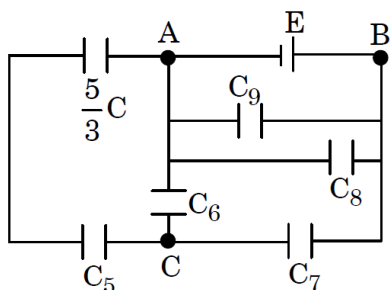
(i) C_2 and C_4 parallel: $C_{24} = 2C$



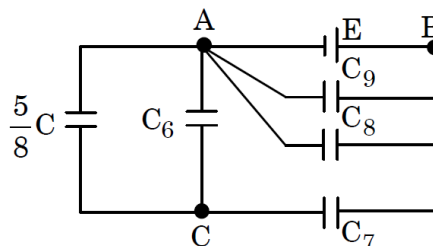
(ii) C_1 and C_{24} series: $\frac{2}{3}C$



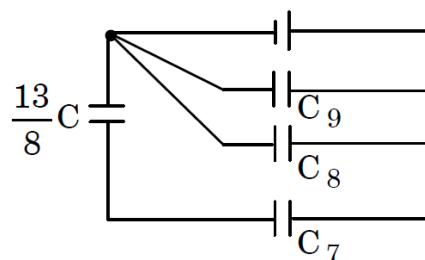
(iii) $\frac{2}{3}C$ and C_3 parallel: $\frac{5}{3}C$



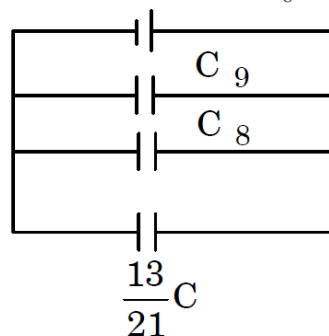
$$(iv) \frac{5}{3}C \text{ and } C_5 \text{ series} \Rightarrow \frac{5}{3} \times \frac{C^2}{\frac{8}{3}C} = \frac{5}{8}C$$



$$(v) \frac{5}{8}C \text{ and } C_6 \text{ parallel: } \frac{13}{8}C$$



$$(vi) \frac{13}{8}C \text{ and } C_7 \text{ series: } \frac{13}{8} \times \frac{C^2}{\frac{21}{8}C} = \frac{13}{21}C$$



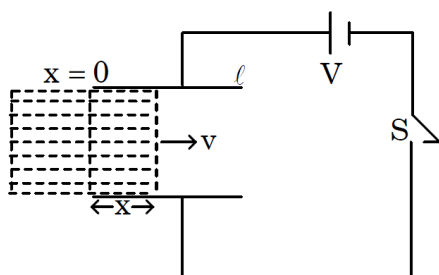
$\therefore$ equivalent capacitance

$$C' = C + C + \frac{13}{21}C = \frac{42 + 13}{21}C = \frac{55}{21}C$$

If $C = 4.2 \text{ mF}$,

$$C' = \frac{55}{21} \times \frac{42}{10} = 11 \text{ mF}$$

6. Ans (A,B)



For dielectric, at $x = 0$, $v = v_0$;

at $x = \ell$, $v = 0$.

$$\Rightarrow v = \frac{v_0}{\ell}(\ell - x)$$

$$\frac{dx}{dt} = \frac{v_0}{\ell}(\ell - x) \text{ or } \frac{dx}{x - \ell} = \frac{v_0}{\ell} dt$$

Integrating, $\ell \ln(x - \ell)$

$$= -\frac{v_0}{\ell}t + \ln c, c - \text{constant}$$

$$\therefore (x - \ell) = ce^{-\frac{v_0}{\ell}t}$$

At $t = 0$, $x = 0$, hence $c = -\ell$.

$$\text{Thus } x = \ell \left(1 - e^{-\frac{v_0}{\ell}t}\right)$$

At an instant t , the effective capacitance is

$$C = \frac{\epsilon_0 K x \ell}{d} + \frac{\epsilon_0 \ell (\ell - x)}{d} = \frac{\epsilon_0 \ell}{d} [x(K - 1) + \ell]$$

$$\text{Charge } q = CV = \frac{\epsilon_0 \ell V}{d} [\ell + x(K - 1)]$$

$$\frac{dq}{dt} = \frac{\epsilon_0 \ell V}{d} (K - 1) \frac{dx}{dt} = \frac{\epsilon_0 \ell V}{d} (K - 1) \frac{v_0}{\ell} (\ell - x)$$

$$\text{Thus, } \frac{dq}{dt} = \frac{\epsilon_0 \ell V}{d} (K - 1) v_0 e^{-\frac{v_0}{\ell}t}$$

$$= \frac{\epsilon_0 V v_0}{d} (\ell - x)$$

$$\text{Energy stored } E = \frac{1}{2} CV^2 = \frac{1}{2} \frac{\epsilon_0 \ell}{d} [x(K - 1) + \ell] V^2$$

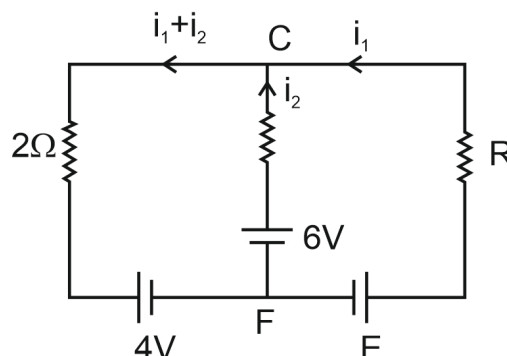
E increases linearly with x , which itself increases exponentially with t , reaching value after a long time. [From (1)]

From equation (2), $\frac{dq}{dt}$ decreases exponentially in time or linearly in x , q increases linearly with x (From (1)).

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (D)



$$-2(i_1 + i_2) - 4 + 6 - i_2 \times 4 = 0$$

$$i_1 + 3i_2 = 1 \quad \dots(i)$$

$$E - i_1 R + i_2 4 - 6 = 0$$

$$E - 6 + i_2 4 = i_1 R \quad \dots(ii)$$

(I) If current in 4Ω is zero then

$$i_2 = 0$$

$$i_1 = 1$$

$$E - 6 + 0 = 1R$$

$$E - 6 = R$$

$$E > 6$$

(II) Current F to C then $i_2 > 0$

$$i_1 = 1 - i_2$$

$$\frac{E - 6 + i_2 \times 4}{1 - 3i_2} = R$$

$$R = +ve$$

$$E - 6 + 4i_2 > 1 - 3i_2$$

$$E > 7 - 7i_2$$

so for PQRS current in 4Ω F to C can be possible same as for B to A possible.

(IV) If current in R is zero then

$$i_1 = 0 \text{ from equation (i)}$$

$$i_2 = \frac{1}{3}$$

put in equation (ii)

$$E - 6 + \frac{1}{3} \times 4 = 0 \times R$$

$$E = 6 - \frac{4}{3} = \frac{14}{3}$$

PART-1 : PHYSICS

SECTION-II

1. **Ans (25.00)**

$$\text{Acceleration } a = A \frac{e^{-2r}}{r^3}, r = (1 \pm 0.05) \mu\text{m}$$

$$\text{Let } X = e^{-2r}$$

$$\frac{\Delta a}{a} = \frac{\Delta X}{X} + 3 \frac{\Delta r}{r}$$

$$\text{Now, } X = e^{-2r}$$

$$\ln X = -2r$$

$$\frac{\Delta X}{X} = -2\Delta r$$

$$\therefore \frac{|\Delta X|}{X} = 2\Delta r = 0.1$$

$$\frac{\Delta a}{a} \% = 10\% + 15\% = 25\%.$$

2. **Ans (0.01)**

$$\frac{\ell \times 4}{\ell \times R_D} = \frac{AD}{DC} = \frac{40}{60} = \frac{2}{3} \text{ or } R_0 = 6$$

$$\frac{R \times 12}{R + 12} = 6 \text{ or } R = 12$$

$$\text{Now, } R = R_0(1 + \alpha \Delta T)$$

$$\text{or } 12 = 6(1 + \alpha \times 100) \text{ or } \alpha = 10^{-2} = 0.01$$

3. **Ans (489.40 to 489.60)**

$$W_{\text{ext}} = \int_0^a \rho a^2 x g dx + \rho a^3 g (h - a)$$

$$= \rho a^3 g \left(h - \frac{a}{2} \right)$$

$$W_{\text{ext}} = 489.51$$

4. **Ans (3.12 to 3.13)**

$$\eta A \frac{dv}{dy} = kx$$

$$\Rightarrow x = \frac{7.5 \times 2 \times 5}{3 \times 80} = 31.25 \text{ cm}$$

5. **Ans (100.00)**

$$C = \frac{3 \times 2}{0.01(3+3)} = 100 \text{ F}$$

7. **Ans (10.00)**

Let current I is flowing in secondary circuit (consisting of 5Ω , X and $5V$). Then, potential difference across 5Ω resistor is proportional to AP if at P , G_1 shows zero deflection.

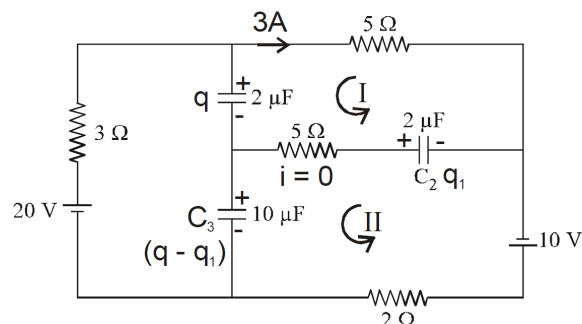
$$\text{i.e., } 5I \propto AP$$

$$\text{Similarly, } (5+x)I \propto AQ$$

$$\Rightarrow \frac{5+X}{5} = \frac{AQ}{AP} = 3$$

$$\Rightarrow X = 10 \Omega$$

8. **Ans (10.00)**



During the steady state the capacitors are fully charged and will behave like open circuit. So the current will flow in outer loop only. The current in the outer loop will be $= \frac{30}{10} = 3$ Amp. in clockwise direction. Charge distribution on the plates of capacitors are as shown in figure.

Applying KVL in loop – I

$$\frac{q}{2} + \frac{q_1}{2} - 15 = 0 \Rightarrow q + q_1 = 30 \quad \text{.....(1)}$$

Applying KVL in loop – II

$$\frac{q - q_1}{10} - 6 + 10 - \frac{q_1}{2} = 0 \quad \text{.....(2)}$$

On solving equation (1) and (2) we get $q_1 = 10 \mu\text{C}$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A,B)**

(a) $P = \alpha V$

$PV^{-1} = \text{constant}$. For a polyprotic process,

$PV^x = \text{constant}$. $x = -1$

$$C = C_V - \frac{R}{x-1}$$

$$C = \frac{R}{\gamma-1} + \frac{R}{2}$$

(b) To calculate work done; $dW = -PdV$

$= -\alpha VdV$. On integrating,

$$W = -\alpha \left(\frac{(4V_0)^2}{2} - \frac{V_0^2}{2} \right)$$

$$W = -15V_0^2 \frac{\alpha}{2}$$

2. **Ans (A,B,D)**

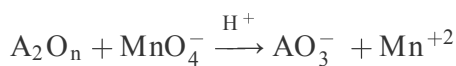
$$\Delta G_{\text{mix}} = RT(n_A \ln \chi_A + n_B \ln \chi_B)$$

$$2 \times 300 \left(1 \ln \frac{1}{3} + 2 \ln \frac{2}{3} \right) = -1145.7 \text{ cal}$$

$$\Delta H = 0 \text{ for ideal solution, thus } \Delta S_{\text{mix}} = \frac{-\Delta G_{\text{mix}}}{T}$$

$$= 3.82 \text{ cal K}^{-1}$$

3. **Ans (A,B,C)**



Eq. of A_2O_n = Eq. of MnO_4

$$1.5 \times 10^{-3} \times (10 - 2n) = 60 \times 0.03 \times 10^{-3} \times 5$$

no. of mole $\times$ (n - factor)

$$\therefore n = 2$$

formula = A_2O_2

Empirical formula = AO

4. **Ans (A,C,D)**

n factor of option B is 6

5. **Ans (B,D)**

Molar conductance increases with dilution

while specific conductance (conductivity)

decreases with dilution. For option C both

equivalent and molar conductance are equal as

n factor is 1.

6. **Ans (B,D)**

$$\text{Rate of adsorption} = K_a P(1 - \theta)$$

$$\text{Rate of desorption} = K_d \theta$$

$$\text{At equilibrium, } K_a P(1 - \theta) = K_d \theta$$

$$\theta = \frac{K_a P}{K_a P + K_d}$$

$$\frac{1}{\theta} = \frac{K_d}{K_a P} + 1$$

for C, slope is minimum, thus C shows the highest extent of adsorption and gas having higher critical temperature shows higher extent of adsorption.

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (D)**

Reaction



$$E = 0.00V - \frac{0.06}{2} \log \frac{[H^+]^2}{pH_2}$$

$$pH_2 = 1 \text{ atm}$$

$$E = -0.059 \log [H^+]$$

$$= 0.059 \times pH$$

$$(a) pH = 7 \text{ (salt does not hydrolyse)} ; E = 0.42 \text{ V}$$

$$(b) pH = 7 - \frac{1}{2} [pK_b + \log c] = 4 ; E = 0.24 \text{ V}$$

$$(c) pH = 7 + \frac{1}{2} [pK_a + \log c] = 10 ; E = 0.6 \text{ V}$$

$$(d) pH = \frac{1}{2} [pK_{a1} + pK_{a2}] = 7 ; E = 0.42 \text{ V}$$

8. **Ans (D)**

Process:

- 1 → 2 Isochoric ($\uparrow$ temp)
2 → 3 Isobaric ($\uparrow$ in $V_d - T$)
3 → 4 Isochoric ($\downarrow$ in T)
4 → 1 Isobaric ($\downarrow$ in $V_d - T$)

Process:

- 1 → 2 Isobaric ($\uparrow$ temp)
2 → 3 Isothermal ($\uparrow V$)
3 → 4 Isochoric ($\downarrow T$)
4 → 1 Isothermal ($\downarrow V$)

Process:

- 1 → 2 Isobaric ($T \uparrow$)
2 → 3 Isochoric ($T \downarrow$)
3 → 4 Isobaric ($\downarrow T$)
4 → 1 Isochoric ($\downarrow V$)

Process:

- 1 → 2 Isothermal ($T \uparrow$)
2 → 3 Isochoric ($V \uparrow$)
3 → 4 Isothermal ($T \downarrow, V \uparrow$)
4 → 1 Isochoric ($V \downarrow$)

9. **Ans (C)**

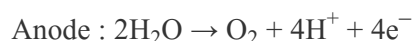
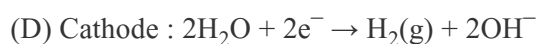
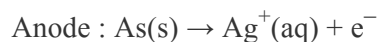
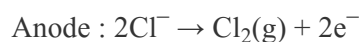
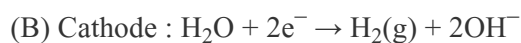
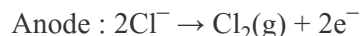
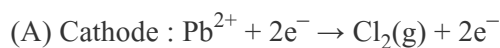
(a) $\Delta H > 0$ as energy is required to break bonds present in protein, $\Delta S > 0$ as albumin gets converted to highly disordered structure in which the molecules aggregate as a disorganized solid, $\Delta G < 0$ as $T\Delta S$ term dominates at higher temperature.

(b) CaCO_3 is being decomposed into gas so $\Delta H > 0$, $\Delta S > 0$, $\Delta G < 0$ as $T\Delta S$ term dominates at higher temperature.

(c) At equilibrium $\Delta G = 0$, fusion of ice is endothermic process and $\Delta S > 0$.

(d) In adsorption of gas in solid $\Delta H < 0$, $\Delta S < 0$ and $\Delta G < 0$ at lower temperature.

10. **Ans (A)**



PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.20)**

$$20 \times N = 20 \times 0.32 \times 5$$

$$N = 1.6 = 0.8M$$

$$\therefore \text{H}_2\text{O}_2 \text{ decomposed } 1 \times 1 - 1 \times 0.2 = 0 \text{ mol}$$

$$\therefore \text{O}_2 \text{ Formed } 0.1 \text{ mol} = 3.2 \text{ gm}$$

2. **Ans (189.00)**

$$\Delta H = \Delta U + \Delta n_g RT$$

$$-380 = \Delta U - \frac{2 \times 2 \times 300}{1000}$$

$$\text{or } -380 = \Delta U - 1.2$$

$$\text{or } \Delta U = -378.8 \text{ K cal mol}^{-1}$$

for 0.5 mole N_2

$$\Delta U = -189.40 \text{ kcal}$$

3. **Ans (2.00)**

$$-w = 3320 = P_{\text{ext}} \Delta V = nR\Delta T$$

$$\Rightarrow \Delta T = \frac{3320}{2 \times 8.3} = 200$$

$$\Rightarrow Q = 11.64 \text{ kJ} = nC_P \Delta T$$

$$\Rightarrow C_P = \frac{11.64 \times 1000}{2 \times 200} = 29.1 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\text{Also, } C_P = \left(\frac{3}{2} + y \right) R \text{ where } y = \text{atomicity}$$

$$\Rightarrow y = \frac{C_P}{R} - \frac{3}{2} = \frac{29.1}{8.3} - 1.5 = 2$$

4. **Ans (3.00)**

$$\Delta T_f = K_f m$$

$$m = \frac{0.93}{1.86} = 0.5$$

$$\Delta T_b = K_b m$$

$$K_b = \frac{0.26}{0.5} = 0.52 \text{ K kg mol}^{-1}$$

$$\text{Similarly } \Delta T_b' = K_b m'$$

$$\Delta T_b' = \frac{K_b \times w \times 1000}{W \times \text{m. wt}}$$

$$\therefore \text{m.wt. of solute} = \frac{K_b \times w \times 1000}{W \times \Delta T_b'}$$

$$= \frac{0.52 \times 7.87 \times 1000}{100 \times 0.44}$$

$$31x = 93$$

$$x = 3$$

5. **Ans (367.50)**

Let the moles of KClO_3 required to be 'x'.

Assuming 50% yield in each step.

$$\text{Moles of } \text{O}_2 \text{ formed in step (i)} = \frac{x}{2}.$$

Mole of each of K_2MnO_4 and Cl_2 formed in

$$(i) = \frac{x}{4}$$

$$\text{Moles of } \text{O}_2 \text{ formed in step (ii)} = \frac{x}{8}$$

$$\text{Total moles of } \text{O}_2 \text{ formed } \frac{x}{2} + \frac{x}{8} = \frac{5x}{8}$$

$$\frac{5x}{8} = \frac{60}{32}$$

$$x = 3$$

6. **Ans (0.25)**

$$(a) \Delta S = 0$$

$$(b) \Delta G = 0$$

(c) $\Delta H > 0$, $\Delta S > 0$, micellisation occurs due to positive change in entropy

(d) Statement corresponds to electro osmosis

(e) Gold sol is hydrophobic sol

(f) True statement

$$(g) E_{\text{Zn}^{2+}/\text{Zn}}^0 = -0.76\text{V} \text{ and}$$

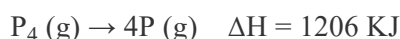
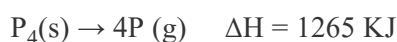
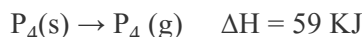
$$E_{\text{H}_2\text{O}/\text{H}_2}^0 = -0.83\text{V} \text{ } E^0 \text{ reaction is -ve}$$

$$(h) K_H \text{ for } \text{CO}_2 = 1.67 \text{ Kbar and } K_H \text{ for } \text{N}_2 = 76.48 \text{ kbar}$$

(i) Van't hoff factor for mohl's salt is higher

(j) Conductivity does not change much initially, after all KI is consumed, conductivity increases

7. **Ans (305.00)**



Average P – P bond enthalpy = 201 KJ

On an average, one P – P bond of tetrahedral is broken and one new P – P bond joining the two tetrahedral units is formed

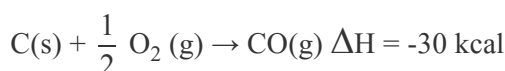
$$201 - x = -104$$

$$\text{Thus, } x = 305 \text{ KJ/mol}$$

8. **Ans (85.71)**



For x mole Carbon, oxygen used up is x mole and heat released is 90x kcal



For (1 – x) mole carbon, oxygen used up is $\frac{1-x}{2}$ and CO formed is 1 – x mole. Heat released in the process is 30(1 – x).

As per question, 1 – x = 0.25, x = 0.75.

Thus, total heat released = 75 cal and moles of oxygen = 0.875.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B,C)**

(A) $f(x) = \{x\} \sin^2 \pi x$ differentiable $\forall x \in \mathbb{R}$

(B) not differentiable at $x = (2n + 1)\pi$

(C) not differentiable at $x = 0, 2, 6, 4$

(D) not differentiable at all integers

2. **Ans (A,C)**

$h(f(x))$ is continuous $\forall x \in \mathbb{R}$ and not differentiable at $x = -1, 1, 2$

3. **Ans (A,C)**

Let $f(x)$ is decreasing $x_2 > x_1 \Rightarrow f(x_2) < f(x_1)$

$$\Rightarrow e^{-f(x_2)} < e^{-f(x_1)} \Rightarrow f(x_2) < f(x_1)$$

$\Rightarrow f(x)$ is increasing

Now $x \rightarrow \infty \Rightarrow f(x) \rightarrow \infty$

$$\frac{\ln x}{f(x)} = 1 + \frac{\ln f(x)}{f(x)} \Rightarrow \lim_{x \rightarrow \infty} \frac{\ln x}{f(x)} = 1$$

4. Ans (A,B,C)

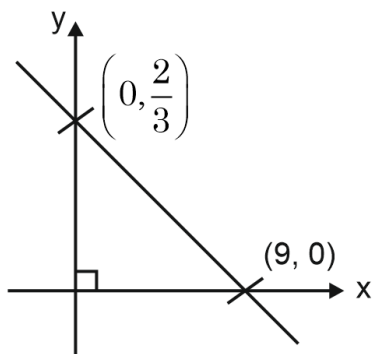
$$\frac{f'(x)}{f(x)} = \left(\frac{-2x-1}{2x(x+1)} + \ln\left(1 + \frac{1}{x}\right) \right) = g(x)$$

$$\Rightarrow g'(x) = \frac{1}{2(x^2+x)^2} > 0$$

5. Ans (A,B)

$$(x^2 - 11)(y + 1) = -4 = -2 \times 2$$

On comparing we get



$$(x_0, y_0) \equiv (3, 1)$$

$$y' = \frac{8x}{(x^2 - 11)^2}$$

$$y'|_{x=3} = 6$$

$$\therefore m_N = \frac{-1}{6}$$

$$\text{Equation of normal } y - 1 = \frac{-1}{6}(x - 3)$$

$$\Rightarrow x + 6y = 9$$

$$\therefore \text{Area} = \frac{1}{2} \times 9 \times \frac{3}{2} = \frac{27}{4}$$

6. Ans (B,C,D)

$F(x)$ is an increasing function and $G(x)$ is a decreasing function

$$(a) F(F(x) - G(x))$$

$$= \underbrace{F'(F(x) - G(x))}_{+ve} \cdot \left(\underbrace{F'(x)}_{+ve} - \underbrace{G'(x)}_{-ve} \right) > 0$$

$$(b) G(F(x) - G(x))$$

$$= \underbrace{G'(F(x) - G(x))}_{-ve} \cdot \left(\underbrace{F'(x)}_{+ve} - \underbrace{G'(x)}_{-ve} \right) < 0$$

$$(c) G'(x) - F'(x) < 0$$

$$(d) G'(F(x)) \cdot F'(x) < 0$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A)

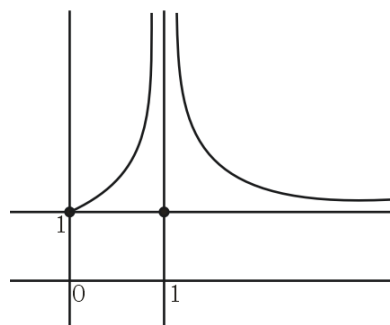
$$F(x) = \frac{(x+1)^2}{(x^2+1)} \times \frac{(x^2+1)^2}{(x^4+1)} \times \frac{(x^4+1)^2}{(x^8+1)} \times \dots$$

$$F(1) = F(0) = 1$$

$$F(x) = \lim_{n \rightarrow \infty} \frac{(x+1)^2(x^2+1)(x^4+1)\dots(x^{2^{n-1}}+1)}{(x^{2^n}+1)}$$

$$F(x) = \lim_{n \rightarrow \infty} \frac{(x+1)^2}{(x^2-1)} \cdot \frac{(x^{2^n}-1)}{(x^{2^n}+1)}$$

$$= \begin{cases} 1 & x = 0 \\ \frac{-(x+1)}{(x-1)} & 0 < x < 1 \\ 1 & x = 1 \\ \frac{x+1}{x-1} & x > 1 \end{cases}$$



8. Ans (C)

$$f_1'(x) = \tan^{-1} x + \frac{x}{1+x^2} \begin{cases} > 0 & \text{if } x > 0 \\ = 0 & \text{if } x = 0 \\ < 0 & \text{if } x < 0 \end{cases}$$

$f_2(x)$ is not continuous at $x = 1$

$$\begin{cases} (x+1) & ; & \text{if } x \in [-2, 0) \\ 0 & ; & \text{if } x = 0 \\ (x+1)e^{-\frac{2}{x}} & ; & \text{if } x \in (0, 2] \end{cases}$$

$$\lim_{x \rightarrow \frac{\pi}{4}} f_4(x) = 1$$

9. **Ans (D)**

Case-I : If $a > 0$, then

$$g \circ f(x) = \begin{cases} x + a + 1; & x < -a \\ (x + a)^2 + 2(b - x - a) + 1; & -a \leq x < 0 \\ (1 - x)^2 + 2(b - 1 + x) + 1; & 0 \leq x < 1 \\ (x - 1)^2 + 2(b - x + 1) + 1; & x \geq 1 \end{cases}$$

$a = 1$ & $b = 0$

Case-II : If $a = 0$, then

$$g(f(x)) = \begin{cases} x + 1 & x < 0 \\ x^2 + 2b & 0 \leq x < 1 \\ (x - 2)^2 + 2b & x \geq 1 \end{cases}$$

$a = 0$; $b = \frac{1}{2}$

$$\left. \frac{d(g(f(x)))}{dx} \right|_{x=4} = (2(x - 1) - 2)|_{x=4} = 4$$

10. **Ans (B)**

$$L = \frac{\lim_{x \rightarrow 0} a \left(x - \frac{x^3}{3!} + \dots \right) + bx \left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots \right) + 3x^2}{\left(\frac{\sin x - x}{x^3} - \frac{x - \tan x}{x^3} \right) x^3}$$

$$L = 6 \lim_{x \rightarrow 0} \frac{x(a + b) + x^2(b + 3) + x^3 \left(\frac{b}{2} - \frac{a}{6} \right)}{x^3}$$

$$L = 6 \left(\frac{b}{2} - \frac{a}{6} \right); a + b = 0 \text{ and } b + 3 = 0$$

$$L = -12, a = 3, b = -3$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (12.50)**

$$f' \left(\frac{a}{x} \right) = \frac{x}{f(x)} \quad \dots(1)$$

$$x \rightarrow \frac{a}{x}$$

$$f'(x) = \frac{a}{xf \left(\frac{a}{x} \right)} \quad \dots(2)$$

$$f''(x) = -\frac{a}{x^2 f \left(\frac{a}{x} \right)} + \frac{a^2 f' \left(\frac{a}{x} \right)}{x^3 \left(f \left(\frac{a}{x} \right) \right)^2} \quad \dots(3)$$

Using equation (1), (2) and (3), we get

$$f''(x) = -\frac{f'(x)}{x} + \frac{(f'(x))^2}{f(x)}$$

$$\Rightarrow xf(x) f''(x) + f(x) f'(x) = x(f'(x))^2$$

$$\Rightarrow \frac{f'(x)}{f(x)} + \frac{xf''(x)}{f(x)} - \frac{x(f'(x))^2}{(f(x))^2} = 0$$

$$\text{Observe } \left(\frac{xf'(x)}{f(x)} \right)' = 0. \text{ So } \frac{xf'(x)}{f(x)} = C$$

$$\Rightarrow \frac{f'(x)}{f(x)} = \frac{C}{x}$$

2. **Ans (2.60)**

Put $n = 1$; $f'(x) = f(x + 1) - f(x)$, $n = 2$;

$$f'(x) = \frac{f(x + 2) - f(x)}{2}$$

$$\text{So, } f'(x) = \frac{f(x + 2) - f(x + 1) + f(x + 1) - f(x)}{2},$$

$$f'(x) = \frac{1}{2} f'(x + 1) + \frac{1}{2} f'(x)$$

$$\Rightarrow f'(x) = f'(x + 1) \quad \forall x \in \mathbb{R} \Rightarrow (f(x + 1) - f(x))'$$

$$= 0 \quad \forall x \in \mathbb{R}$$

$$\Rightarrow f(x + 1) - f(x) = c \text{ for a constant } c \in \mathbb{R}$$

$$\Rightarrow f'(x) = c \Rightarrow f(x) = cx + d$$

3. **Ans (0.20)**

$$f'(x) = k(x - 1)(x - 3)(x - 2)^2 \text{ and } k > 0$$

$$\Rightarrow f(x) = k \int (x^2 - 4x + 3)(x^2 - 4x + 4) dx$$

$$k \int x^4 + 16x^2 - 8x^3 - 7(x^2 - 4x) + 12 dx$$

$$= k \left(\frac{x^5}{5} - 2x^4 + \frac{23x^3}{3} - 14x^2 + 12x \right) + c$$

$$\therefore f(0) = 2 \Rightarrow c = 2 \therefore f(1) = \frac{88}{15}$$

$$\Rightarrow \frac{88}{15} = k \left(\frac{1}{5} - 2 + \frac{23}{3} - 14 + 12 \right) + c$$

$$\Rightarrow \frac{58}{15} = k \times \frac{58}{15} \Rightarrow k = 1 \therefore a_5 = \frac{1}{5}$$

4. **Ans (0.07)**

$$\text{Let } t^3 = x \Rightarrow \lim_{t \rightarrow 0} \frac{\sin x - x \cos x + x^5}{x(e^{3x} - 1 - 3x)}$$

Now use expansions of $\sin x$, $\cos x$ and e^{3x} .

5. **Ans (2.00)**

For $1 \leq k \leq n$

$$\left(1 + \frac{x}{n} \right)^{k/n} \leq \left(1 + \frac{kx}{n^2} \right) \leq \left(1 + \frac{x}{n^2} \right)^k$$

$$\left(1 + \frac{x}{n} \right)^{\frac{n+1}{2}} \leq P_n \leq \left(1 + \frac{x}{n^2} \right)^{\frac{n(n+1)}{2}}$$

$$e^{x/2} \leq \lim_{n \rightarrow \infty} P_n \leq e^{x/2} \Rightarrow f(x) = (e)^{x/2}$$

6. Ans (1275.00)

$$f(x) = \prod_{n=1}^{50} (x-n)^{n(51-n)}$$

$$\ln(f(x)) = \prod_{n=1}^{50} n(51-n) \ln(x-n)$$

Differentiation both side w.r.t. x

$$\frac{f'(x)}{f(x)} = \sum_{n=1}^{50} \frac{n(51-n)}{x-n}$$

$$\Rightarrow \frac{f'(51)}{f(51)} = 1275$$

7. Ans (7.00)

$$\frac{f(3x+3y)}{\sin(2x+2y)} = \frac{f(3x-3y)}{\sin(2x-2y)} \quad \forall x, y \in \mathbb{R}$$

$$\Rightarrow f(x) = k \sin\left(\frac{2x}{3}\right)$$

$$\therefore f'(0) = \frac{2}{3} \sin\left(\frac{2x}{3}\right) \Rightarrow f'\left(\frac{\pi}{2}\right) = \frac{3\sqrt{3}}{4}$$

$$\text{Equation of tangent is } y - \frac{3\sqrt{3}}{4} = \frac{1}{2} \left(x - \frac{\pi}{2}\right)$$

8. Ans (0.00)

$$g''(x) = - \left(\frac{f''(x)}{(f'(x))^3} \right)_{x=\frac{3\pi}{2}}$$

$$f'(x) = 6(2x-3\pi)^2 + \frac{4}{3} - \sin x$$

$$f'\left(\frac{3\pi}{2}\right) = \frac{7}{3}$$

$$f''(x) = 24(2x-3\pi) - \cos x$$

$$f''\left(\frac{3\pi}{2}\right) = 0$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

PART TEST

06-12-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TEAS, TOAS, TAAS, TNAS, I(A) & LIVE-I(A)] (SCORE-I)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	C	D	B	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	C	A,C	B,C	B	A,B,D	A,B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	1	5	5	1	5	6	8	4

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	A	A	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	A,C,D	A,B,C	B,D	A,B,D	A,B,C	B,C		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	4	1	6	2	6	2	3	4

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	B	A	A	D				
SECTION-I (ii)	Q.	5	6	7	8	9	10		
	A.	B,C,D	B,C	A,B,C	A,B,C	A,B,C	A,B,D		
SECTION-III	Q.	1	2	3	4	5	6	7	8
	A.	2	1	5	8	0	9	0	9

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

$$V_B = 0 - 2 = -2V, V_D = 0 + 5 - 10 = -5V$$

4. Ans (B)

$$1A \times \frac{r_1}{\frac{Rr_2}{R+r_2} + R + r_1} = 5mA \quad \dots(i)$$

$$\text{and } 5mA \times \frac{r_2}{R + r_2} = 2mA \quad \dots(ii)$$

$$r_2 = \frac{2R}{3}$$

solving equation (i)

$$r_1 = \frac{7R}{995}$$

PART-1 : PHYSICS

SECTION-I (ii)

5. Ans (C)

As the tank allowed to fall vertically, so buoyant force becomes zero.

6. Ans (A,C)

$$P = \frac{500}{100 \times 10^{-4}} = 5 \times 10^4 \text{ N/m}^2$$

Using Bernoulli's equation, we have

$$P + 0 + 0 = P_a + \frac{1}{2}\rho v^2 + \rho gh$$

$$= 1 \times 10^5 + \frac{1}{2} \times 10^3 \times 10^2 + 10^3 \times 10 \times 0.50$$

$$= 1.55 \times 10^5 \text{ N/m}^2$$

If F is the force applied then

$$P_a + \frac{F}{A} = 1.55 \times 10^5$$

$$\text{or } 1 \times 10^5 + \frac{F}{A} = 1.55 \times 10^5$$

$$\therefore F = 0.55 \times 10^5 \times 100 \times 10^{-4} = 550 \text{ N}$$

7. Ans (B,C)

$$E = \frac{4Q}{2A\epsilon_0} + \frac{2Q}{2A\epsilon_0}$$

$$E = \frac{3Q}{Cd} ; Ed = \frac{3Q}{C} = V$$

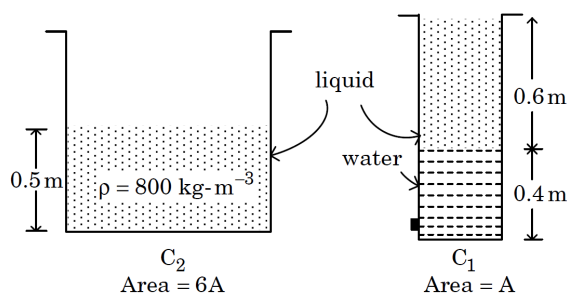
$$\text{Energy} = \frac{\left[4Q - \left(\frac{4Q-2Q}{2}\right)\right]^2}{2C} = \frac{9Q^2}{2C}$$

$$F = \frac{4Q}{2A\epsilon_0} \times 2Q = \frac{4Q^2}{Cd}$$

8. Ans (B)

(A) When C_1 is immersed in C_2 , height of the level of liquid in C_2 is h (say)

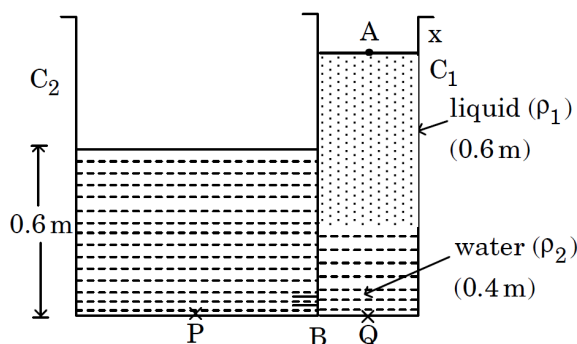
Before



$$6A \times 0.5 = 5A \times h$$

$$h = 0.6 \text{ m}$$

After



Let v be the speed with which water comes out from the orifice. Applying Bernoulli's theorem, for stream line flow at A and B,

$$p_A + 0 + (0.6 \rho_1 + 0.4 \rho_2)g = p_B + \frac{1}{2} \rho_2 v^2$$

assuming level A is coming down very slowly

(area of orifice $a \ll A$)

$$p_A = p_0 \text{ (atmospheric pressure)}$$

$$p_B = p_0 + 0.6 \rho_1 g$$

$$\frac{1}{2} \rho_2 v^2 = 0.4 \rho_2 g$$

$$v^2 = 0.8 \times 10 = 8 \text{ so } v = 2\sqrt{2} \text{ ms}^{-1}$$

(B) Let x be the distance through which liquid level in C_1 falls. Then, height of water column in $C_1 = (0.4 - x) \text{ m}$

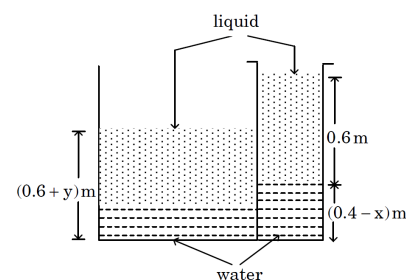
Let y be the height to which liquid in C_2 is increased.

Volume of water decreased in $C_1 = \text{Volume of water increased in } C_2$

$$\therefore Ax = 5A \times y \text{ or } x = 5y$$

Further, pressure at P (in C_2) = Pressure at Q (in C_1)

$$p_0 + (0.6 \times 800 + y \times 1000)g = p_0 + (0.6 \times 800 + (0.4 - x) \times 1000)g$$



$$y = 0.4 - x$$

$$x = 0.4 - y = 0.4 - \frac{x}{5}$$

$$5x = 2 - x$$

$$x = \frac{2}{6} = \frac{1}{3} \text{ m}$$

$\therefore$ length of water column

$$\text{in } C_1 = \left(0.4 - \frac{1}{3}\right) \text{ m}$$

(C) Pressure exerted at the bottom of C_2 in steady state is

$$p = p_0 + (0.6 \times 800 + y \times 1000)g$$

$$y = \frac{x}{5} = \frac{1}{15} \text{ m,}$$

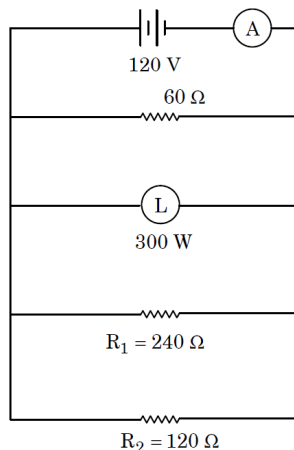
$$\therefore p = p_0 + \left(480 + \frac{1000}{15}\right) 10$$

$$p_0 + \left(\frac{1440 + 200}{3}\right) 10$$

$$p_0 + \frac{16400}{3} = p_0 + 5466.6 = 1 \times 10^5 + 0.055 \times$$

$$10^5 = 1.06 \times 10^5 \text{ Nm}^{-2}.$$

9. Ans (A,B,D)



The given circuit can be redrawn as follows.

$$\text{Current through } R_1 = \frac{120}{240} = \frac{1}{2} \text{ A}$$

$$\therefore \text{ power dissipated in } R_1 = \frac{1}{4} \times 240$$

$$= 60 \text{ W}$$

$$\text{Power dissipated in } R_2 = \left(\frac{120}{120} \right)^2 \times 120$$

$$= 120 \text{ W}$$

Current in the 300 W lamp

$$= \frac{300}{120} = \frac{5}{2} \text{ A}$$

Resistance in the 300 W lamp

$$= \frac{120 \times 120}{300} = 48 \Omega$$

$$\text{Effective resistance } \frac{1}{R} = \frac{1}{48} + \frac{1}{60} + \frac{1}{240} + \frac{1}{120}$$

$$\therefore R = \frac{240}{12} = 20 \Omega$$

$$\text{Current passing through A is } \frac{120}{20} = 6 \text{ A.}$$

10. Ans (A,B,D)

Potential difference across resistance = potential difference across the terminals of the battery.

$$\text{So, } V = E - ir$$

This is an equation of a straight line comparing this with the given graph we can see that

$$E = 16 \text{ V and } r = 4 \Omega$$

$$\text{Also } i_{\max} = \frac{E}{r} = 4 \text{ A when external resistance}$$

$$R = 0.$$

For max power, $R = r = 4 \Omega$

$$P_{\max} = \frac{8^2}{4} = 16 \text{ J}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (1)

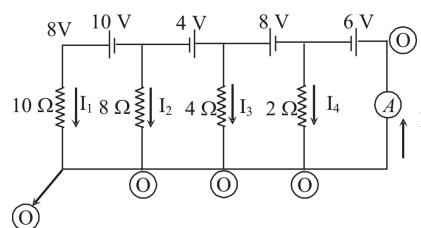
$$F = \frac{2T A}{d}$$

Here, $d = h$, $A h \rho = m$

$$\therefore A = \frac{m}{h \rho}$$

$$\therefore F = \frac{2T \left(\frac{m}{h \rho} \right)}{h} = \frac{2T m}{\rho h^2}$$

2. Ans (5)



$$I = I_1 + I_2 + I_3 + I_4 = \frac{8}{10} - \frac{2}{8} + \frac{2}{4} - \frac{6}{2}$$

$$\frac{32 - 10 + 20 - 120}{40} = \frac{-78}{40} = -1.95 \text{ A}$$

$$I = 0.39 \times 5$$

3. Ans (5)

Total initial energy,

$$E_i = \frac{1}{2} C V^2 + \frac{1}{2} C V^2 = C V^2$$

Total final energy,

$$E_f = \frac{3}{2} C V^2 + \frac{C V^2}{6} = \frac{5}{3} C V^2$$

$$\therefore \frac{E_i}{E_f} = \frac{C V^2}{(5/3) C V^2} = \frac{3}{5}$$

$$N = 5$$

4. Ans (1)

Let maximum charge be Q . finally, charge will decrease from Q to q_0 during discharge and then increase from q_0 to Q during charging.

$$q = Q e^{-t/\tau}$$

$$q_0 = \frac{Q}{e}$$

during discharging,

$$q = A + B e^{-t/\tau}$$

$$A + B = \frac{Q}{e}$$

$$B = \left(\frac{Q}{e} - C E \right)$$

$$Q = C E + \left(\frac{Q}{e} - C E \right) \frac{1}{e}$$

$$Q = \frac{e}{e+1} \cdot E C$$

5. Ans (5)

Charge on the capacitor,

$$Q = CV = 20 \times 10^{-9} \times 2500 = 5 \times 10^{-5} \text{C}$$

Surface charge density,

$$\sigma_s = \frac{Q}{A} \text{ [A is the plate area]}$$

and electric field strength between the plates,

$$E = \frac{\sigma_s}{K\epsilon_0} = \frac{Q}{KA\epsilon_0}$$

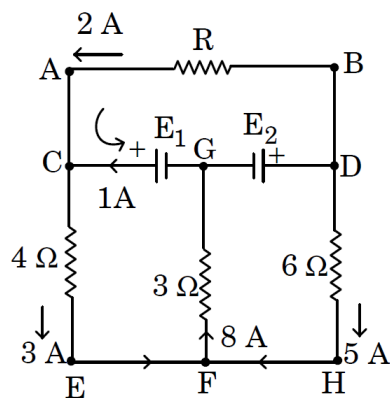
$$\text{Using, } J = \sigma E = \frac{E}{\rho}$$

[here σ is conductivity and ρ the resistivity]

$$\text{Current density, } J = \frac{Q}{KA\epsilon_0\rho}$$

$$\begin{aligned} \text{and current} &= JA = \frac{Q}{K\epsilon_0\rho} \\ &= \frac{5 \times 10^{-5}}{6 \times 8.85 \times 10^{-12} \times 2 \times 10^{11}} \\ &= 4.7 \times 10^{-6} \text{ A} = 4.7 \mu\text{A} \end{aligned}$$

7. Ans (8)



By Kirchhoff's I law current through 3Ω is $5 + 3 = 8 \text{ A}$

8. Ans (4)

$$f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$$

$$\therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{(\ell + 1.2r)}$$

$$\begin{aligned} \text{Here } \frac{\Delta f}{f} \times 100 &= \frac{\Delta(\ell + 1.2\Delta r)}{(\ell + 1.2r)} \times 100 \\ &= 0.16 \end{aligned}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A)

Vant Hoff's equation

$$\frac{d \ln k}{dT} = \frac{\Delta H^\circ}{RT^2}$$

$$\ln K = \frac{-\Delta H^\circ}{RT} + \text{constant}$$

For endothermic reaction, on increasing $1/T$ more fall in K for more endothermic reaction

For exothermic reaction, on increasing $1/T$ more increase in K for more exothermic reaction

$$\Delta H_4 > \Delta H_3 > \Delta H_2 > \Delta H_1$$

ΔH_2 & ΔH_1 are -ve and ΔH_3 & ΔH_4 are +ve.

2. Ans (A)

$$\text{For first cell } x = 0 - \frac{0.059}{1} \log \frac{[\text{Ag}^+]_{\text{anode}}}{[\text{Ag}^+]_{\text{cathode}}}$$

Let a mole/L be the solubility of Ag_2CrO_4

$$K_{sp} = (2a)^2 a$$

$$= 4a^3$$

$$\text{then } [\text{Ag}^+] = 2a$$

$$= 2 \left(\frac{K_{sp}}{4} \right)^{1/3}$$

$$X = 0 - \frac{0.059}{1} \log \frac{2 \times \left(\frac{p}{4} \right)^{1/3}}{0.1}$$

similarly for 2nd cell reaction

$$x = 0 - \frac{0.059}{1} \log \frac{q^{1/2}}{0.1}$$

for 3rd cell reaction

$$x = 0 - \frac{0.059}{1} \log \frac{1}{2 \times \left(\frac{s}{4} \right)^{1/3}}$$

At equilibrium $E_{\text{cell}} = x = 0$, so

$$\frac{2 \left(\frac{p}{4} \right)^{1/3}}{0.1} = \frac{q^{1/2}}{0.1} = \frac{3 \left(\frac{r}{27} \right)^{1/4}}{0.1} = \frac{1}{2 \left(\frac{s}{4} \right)^{1/3}} = 1$$

$$p = 5 \times 10^{-4}$$

$$q = 10^{-2}$$

$$r = \frac{10^{-4}}{3}$$

$$s = 0.5$$

$$s > q > p > r$$

3. **Ans (A)**

100 ml of 0.5 M CH_3COOH contains = 0.05 mole of CH_3COOH

After adsorption, CH_3COOH present = 0.049 mole

$\therefore$ Acetic acid adsorbed by 1 g charcoal = 0.001 mole

$$= 6.02 \times 10^{20} \text{ molecules}$$

Surface area of 1g of charcoal

$$= 3.01 \times 10^2 \text{ m}^2$$

$\therefore$ Surface area of charcoal adsorbed by each molecule

$$= \frac{3.01 \times 10^2}{6.02 \times 10^{20}} \text{ m}^2$$

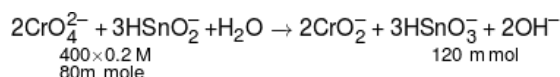
$$= 5 \times 10^{-19} \text{ m}^2$$

PART-2 : CHEMISTRY

SECTION-I (ii)

5. **Ans (A,C,D)**

Balanced Equation



here limiting reagent is CrO_4^{2-}

At the end of reaction the concentration of

$$\text{CrO}_2^- \text{ ions} = \frac{80}{900} = 0.0888 \approx 0.09 \text{ M}$$

At the end of reaction, then concentration of

$$\text{HSnO}_3^- \text{ ions} = \frac{120}{900} = 0.1333 \approx 0.13 \text{ M}$$

6. **Ans (A,B,C)**

For adiabatic process, heat change = 0

$$\Delta E = -P_{\text{ext}} \Delta V$$

$$C_V \Delta T = -P_{\text{ext}} \Delta V$$

$$C_V \Delta T = -2P_1 (V_2 - V_1) = -RT_2 + 2RT_1$$

$$= -R(T_2 - T_1) + RT_1$$

$$\Rightarrow \frac{3}{2} R \Delta T = -R \Delta T + RT_1$$

$$\Rightarrow \frac{5}{2} R \Delta T = RT_1$$

$$\Delta T = \frac{2}{5} T_1$$

$$\text{Again } C_V \Delta T = -2P_1 \Delta V$$

$$C_V \frac{2}{5} T_1 = -2P_1 \Delta V \Rightarrow \Delta V = \frac{-3}{10} \frac{RT_1}{P_1}$$

$P_1 V_1^\gamma = P_2 V_2^\gamma$ is true only for reversible adiabatic process.

7. **Ans (B,D)**

$$(A) M = \frac{\text{"v" strength}}{11.35} \Rightarrow \frac{34.05}{11.35} = 3$$

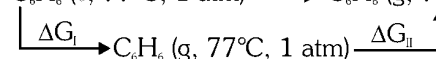
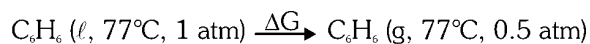
$$(B) \% \frac{w}{v} = \frac{3 \times 34}{1000} \times 100 = 10.2$$

$$(C) \text{Mol fraction of } \text{H}_2\text{O}_2 = \frac{3}{3 + 57} = 0.05$$

$$(D) m_{\text{H}_2\text{O}_2} = \frac{3000}{1128 - 3 \times 34} = 2.924$$

$\therefore$ option (B) and (D)

8. **Ans (A,B,D)**



$$\Delta G = \Delta G_I + \Delta G_{II}$$

$$\Delta G_I = 0, \Delta G_{II} = nRT \ln \frac{P_2}{P_1}$$

$$= - \left[\frac{3 \times 8 \times 350}{10^3} \ln \frac{0.5}{1} \right] \text{ kJ} = -5.88 \text{ kJ}$$

PART-2 : CHEMISTRY

SECTION-III

1. **Ans (4)**

$$\frac{1}{\lambda_m} = \frac{1}{\lambda_m^0} + \frac{\lambda_m C}{K_a (\lambda_m^0)^2}$$

$$C \lambda_m = 0.01 \frac{S}{\text{decimeter}} = 0.01 \frac{\text{mole}}{L} \times S \text{ cm}^2 \text{ mole}^{-1}$$

$$\text{Slope} = \frac{1}{K_a (\lambda_m^0)^2} = 1$$

$$\frac{1}{\lambda_m^0} = 0.01 = \text{intercept on y axis}$$

$$K_a = \left(\frac{1}{\lambda_m^0} \right)^2 = 10^{-4}$$

$$\text{p}K_a = 4$$

2. **Ans (1)**

For the first part, millimole of $\text{Na}_2\text{S}_2\text{O}_3$ consumed
 $= 15 \times 0.4 = 6$

$\text{I}_2 + 2\text{Na}_2\text{S}_2\text{O}_3 \rightarrow 2\text{NaI} + \text{Na}_2\text{S}_4\text{O}_6$ millimoles of
 I_2 consumed = 3 millimoles (i)

For the second part, the reaction is $3\text{I}_2 + 6\text{NaOH}$
 $\rightarrow 5\text{NaI} + \text{NaIO}_3 + 3 \text{H}_2\text{O}$ reacted with I_2 so,
 millimoles of NaOH
 $= (100 \times 0.3) - (10 \times 0.3 \times 2) = 30 - 6 = 24$
 millimoles.

$$\text{Millimoles of } \text{I}_2 \text{ consumed} = \frac{24}{2} = 12 \text{ millimoles,}$$

→ (ii)

Total millimoles of I_2 consumed in reaction
 $= (\text{i}) + (\text{ii}) = 3 + 12 = 15 \text{ millimoles}$

∴ molarity of $\text{I}_2 = \frac{15}{150} = 0.1\text{M}$ "10 times initial
 molarity of $\text{I}_2 = 0.1 \times 10 = 1$

3. **Ans (6)**

$$\% \text{N} = \frac{1.4 \times (5.0 - 2.5)}{0.303} = 11.55$$

In compound $\text{C}_w\text{H}_x\text{N}_y\text{O}_z$

$$\% \text{C} = 69.4 = \frac{12w}{121} \times 100; w = 6.99 \simeq 7$$

$$\% \text{H} = 5.8 = \frac{x}{121} \times 100; x = 7$$

$$\% \text{N} = 11.55 = \frac{14y}{121} \times 100; y = 0.998 \simeq 1$$

$$\% \text{O} = 13.25 = \frac{16z}{121} \times 100; z = 1$$

So, the compound is $\text{C}_7\text{H}_7\text{NO}$.

$$\text{now, } \frac{w+x-y-z}{2} = \frac{1+7-1-1}{2} = 6$$

4. **Ans (2)**

Basis of calculation = $10^6 \text{ g hard water}$

$$\text{MgSO}_4 = 0.00012\% = \frac{0.00012}{100} \times 10^6 \text{ g} = 1.2 \text{ g}$$

$$= \frac{1.2}{120} \text{ mole} = 0.01 \text{ mole}$$

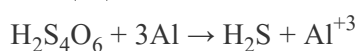
$$\text{CaCl}_2 = 0.000111\% = \frac{0.000111}{100} \times 10^6 \text{ g}$$

$$= 1.11 \text{ g} = \frac{1.11}{111} \text{ mole} = 0.01 \text{ mole}$$

$$\text{CaCO}_3 = 0.01 + 0.01 = 0.02 \text{ mole}$$

$$= 0.02 \times 100 \text{ g} = 2\text{g} = \boxed{2 \text{ PPM}}$$

5. **Ans (6)**



n - factor for Al = 3

n - factor for $\text{H}_2\text{S}_4\text{O}_6 = 18$

$$\therefore \text{Coefficient of Al} = \frac{18}{3} = 6$$

6. **Ans (2)**

$$P = K_H \cdot x$$

$$\Rightarrow x = \frac{P}{K_H} = \frac{2.5 \times 10^5}{1.67 \times 10^8} = 1.5 \times 10^{-3}$$

$$\text{Now, } 1.5 \times 10^{-3} = \frac{n_{\text{CO}_2}}{n_{\text{CO}_2} + \frac{550}{18}}$$

$$n_{\text{CO}_2} + \frac{550}{18} \approx \frac{550}{18}$$

$$\Rightarrow 1.5 \times 10^{-3} = \frac{n_{\text{CO}_2}}{\frac{550}{18}}$$

$$\Rightarrow n_{\text{CO}_2} = \frac{1.5 \times 10^{-3} \times 550}{18}$$

Mass of CO_2

$$= n_{\text{CO}_2} = \frac{1.5 \times 10^{-3} \times 550 \times 44}{18} \approx 2 \text{ g}$$

7. **Ans (3)**

m.eq. of $\text{KMnO}_4 = \text{m.eq. of } \text{NH}_2\text{OH}$

$$\Rightarrow 10 \times 0.02 \times 5 = 50 \times N$$

$$\Rightarrow N = \frac{1}{50}$$

$$\text{Eq. of mass of } \text{NH}_2\text{OH} = \frac{M}{2} = \frac{33}{2} = 16.6$$

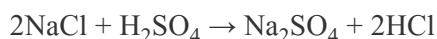
$$\frac{1}{50} = \frac{x}{16.5}$$

$$\Rightarrow x = \frac{16.5}{50} \text{ g}$$

$$\text{Mass in 10 mL of original solution} = \frac{16.5}{50} \text{ g}$$

$$\text{Mass in 91 mL of original solution} = 3\text{g}$$

8. Ans (4)



$$\therefore a = 1, b = 5, c = 2, d = 2, e = 2$$

$$(a + b + c) - (d + e) = (1 + 5 + 2) - (2 + 2)$$

$$= 4$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (B)

$$f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right)$$

$$+ a \geq 0 \quad \forall x > 0 \text{ and}$$

$$f(x) = \left(x^3 + \frac{1}{x^3}\right) - 6\left(x^2 + \frac{1}{x^2}\right) + 12\left(x + \frac{1}{x}\right)$$

$$+ a \leq 0 \quad \forall x < 0$$

$$\text{Let } x + \frac{1}{x} = t$$

$$\text{So, } g(t) = t^3 - 6t^2 + 9t + 12 + a \geq 0; t \geq 2$$

$$g'(t) = 3(t-1)(t-3) \Rightarrow g(3) \geq 0 \text{ and } g(t) \leq 0$$

$$\forall t \leq -2 \Rightarrow -12 \leq a \leq 38$$

2. Ans (A)

$$f(x) = \sin^3 x + \lambda \sin^2 x$$

$$f'(x) = 3 \sin^2 x \cos x + 2\lambda \sin x \cos x$$

for exactly one minimum and one maximum

$$f'(x) = 0 \text{ at two points in } \frac{-\pi}{2} < x < \frac{\pi}{2}$$

$$\text{therefore } f'(x) = 0 \Rightarrow \cos x = 0 \text{ or } \sin x (3 \sin x + 2\lambda)$$

$$= 0 \Rightarrow \sin x = \frac{-2\lambda}{3} \text{ or } \sin x = 0, \cos x \neq 0$$

$$\Rightarrow -1 < \frac{-2\lambda}{3} < 1$$

$$\Rightarrow -\frac{3}{2} < \lambda < \frac{3}{2} \text{ but } \lambda > 0 \Rightarrow 0 < \lambda < \frac{3}{2}$$

3. Ans (A)

We have L

$$= \lim_{x \rightarrow 2} \frac{\sqrt{(x+7)} - 3\sqrt{(2x-3)}}{\sqrt[3]{(x+6)} - 2\sqrt[3]{3x-5}} \left(\frac{0}{0} \text{ form}\right)$$

Let $x - 2 = t$ such that when $x \rightarrow 2$, $t \rightarrow 0$. Then

$$L = \lim_{t \rightarrow 0} \frac{\frac{1}{(t+9)^{\frac{2}{3}}} - 3\frac{1}{(2t+1)^{\frac{2}{3}}}}{\frac{1}{(t+8)^{\frac{1}{3}}} - 2\frac{1}{(3t+1)^{\frac{1}{3}}}} \left(\frac{0}{0} \text{ form}\right)$$

$$= \frac{3}{2} \lim_{t \rightarrow 0} \frac{\left(1 + \frac{t}{9}\right)^{\frac{1}{2}} - (2t+1)^{\frac{1}{2}}}{\left(1 + \frac{t}{8}\right)^{\frac{1}{3}} - (3t+1)^{\frac{1}{3}}} \left(\frac{0}{0} \text{ form}\right)$$

$$= \frac{3}{2} \lim_{t \rightarrow 0} \frac{\frac{1}{2} \frac{t}{9} - (2t)^{\frac{1}{2}}}{\frac{1}{3} \frac{t}{8} - (3t)^{\frac{1}{3}}} = \frac{3}{2} \frac{\frac{1}{18} - 1}{\frac{1}{24} - 1} = \frac{34}{23}$$

4. Ans (D)

$$f(0^-) = 0, f(0^+) = 1, f(0) = 1$$

$\Rightarrow$ discontinuous at $x = 0$

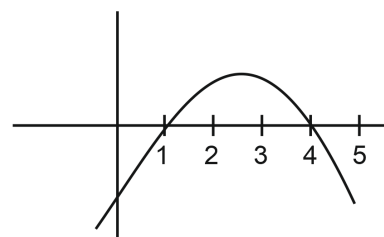
$$f(1^-) = 1, f(1) = 1, f(1^+) = 0$$

$\Rightarrow$ discontinuous at $x = 1$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. Ans (B,C,D)



$$f(x) = \begin{cases} -(x-1)(x-5) & x \leq 3 \\ -4(x-2)(x-4) & x > 3 \end{cases}$$

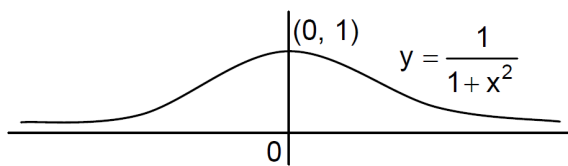
Lowest y intercept will be of tangent made at (1, 0) and highest y intercept will be of tangent

made at (4, 0). Tangent at (1, 0) is $y = 4x - 4$

$\therefore m_2 = 4, b_2 = -4$ and tangent at (4, 0) is

$y = -8x + 32 \therefore m_1 = -8, b_1 = 32$

6. Ans (B,C)



$f(x) > 0$ and $f'(x) > 0$ for $x < 0$

$f'(x) < 0$ for $x > 0$

$$\text{Now } g(x) = \frac{1}{f\left(\frac{1}{x}\right)}$$

$$g'(x) = \frac{-1}{f^2\left(\frac{1}{x}\right)} \times f'\left(\frac{1}{x}\right) \times \left(\frac{-1}{x^2}\right)$$

So for $x < 0 \Rightarrow g'(x) > 0$

$x > 0 \Rightarrow g'(x) < 0$

7. Ans (A,B,C)

Since, f attains its maximum value at some $c \in (0, 4)$

$$\therefore f'(c) = 0$$

Now, in $y = f(x)$ using LMVT in $[0, c]$ and $[c, 4]$

$$|f''(d)| = \left| \frac{f'(c) - f'(0)}{c} \right| \leq 5$$

$$\Rightarrow |f'(0)| \leq 5c$$

$$\text{Again, } |f''(d)| = \left| \frac{f'(4) - f'(c)}{4 - c} \right| \geq 5$$

$$\Rightarrow |f'(4)| \leq 5(4 - c)$$

$$\text{Eqn. (1) + eqn. (2)} \Rightarrow |f'(0)| + |f'(4)| \leq 20$$

8. Ans (A,B,C)

$$F(x) = f(x) \cdot g(x)$$

Differentiate both sides w.r.t. x

$$F'(x) = f'(x)g(x) + g'(x)f(x) \dots \dots \dots (i)$$

$$F'(x) = f'(x)g'(x) \left[\frac{g(x)}{g'(x)} + \frac{f(x)}{f'(x)} \right]$$

$$F' = c \left[\frac{g}{g'} + \frac{f}{f'} \right]$$

Again differentiate equation (i), w.r.t. x , we get

$$F''(x) = f''(x)g(x) + g'(x)f'(x) + g''(x)f(x) + f'(x)g'(x)$$

$$\Rightarrow F''(x) = f''(x)g(x) + g''(x)f(x) + 2g'(x) \cdot f'(x)$$

.....(ii)

Divide $F(x) = f(x) \cdot g(x)$ on both sides

$$\Rightarrow \frac{F''(x)}{F(x)} = \frac{f''(x)g(x)}{F(x)} + \frac{g''(x)f(x)}{F(x)} + \frac{2c}{F(x)}$$

$$\Rightarrow \frac{F''(x)}{F(x)} = \frac{f''(x)}{f(x)} + \frac{g''(x)}{g(x)} + \frac{2c}{f(x)g(x)}$$

$$\Rightarrow \frac{F''}{F} = \frac{f''}{f} + \frac{g''}{g} + \frac{2c}{fg}$$

Again differentiate equation (ii)

$$F'''(x) = f''(x)g'(x) + f'''(x)g(x) + g''(x)f'(x) + g'''(x)f(x) + 0\{g'(x)f'(x) = c\}$$

$$\Rightarrow F'''(x) = f'''(x)g(x) + g'''(x)f(x) + f''(x)g'(x) + g''(x)f'(x)$$

.....(iii)

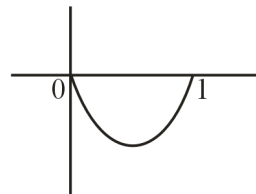
$$f'(x)g'(x) = c$$

$$\text{Differentiate both sides } f'(x)g''(x) + g'(x)f''(x) = 0$$

Putting this in equation (iii), we get

$$\frac{F'''}{F} = \frac{f'''}{f} + \frac{g'''}{g}$$

9. Ans (A,B,C)



$$(f(x)e^x)'' \geq 0$$

So, $f(x)e^x$ is convex

$$\text{So, } f(x)e^x \leq \max[f(0), f(1)] = 0$$

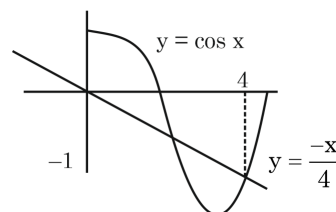
$$\therefore f(x) \leq 0 \quad \forall x \in [0, 1]$$

10. Ans (A,B,D)

$$(a) f(x) = \ln(1+x) - x + \frac{x^2}{2}$$

$$f'(x) = \frac{1}{1+x} - 1 + x = \frac{x^2}{1+x}$$

$$f(x) = 0, f'(x) > 0 \quad \forall x \in (0, \infty) \Rightarrow f(x) > 0$$



$$(b) f(x) = \ln(1+x) - \frac{x}{4}$$

$$f'(x) = \frac{1}{1+x} - \frac{1}{4} = \frac{x}{(1+x)^2}$$

$$f(0) = 0, f'(x) > 0 \quad \forall x \in (0, \infty) \Rightarrow f(x) > 0$$

$$(c) y = \cos x, y = -\frac{x}{4}$$

by graph it is clear that

$$\cos x > -\frac{x}{4} \text{ is not true } \forall x \in (0, \infty)$$

$$(d) f(x) = x + 1 - 2\tan^{-1}x$$

$$f(0) = 1$$

$$f'(x) = \frac{x^2 - 1}{1 + x^2} = \frac{(x-1)(x+1)}{x^2 + 1}$$

at $x = 1$ $f(x)$ has minima

$$f(1) = 2 \left(1 - \frac{\pi}{4}\right) > 0 \Rightarrow f(x) > 0 \forall x \in (0, \infty)$$

PART-3 : MATHEMATICS

SECTION-III

1. **Ans (2)**

$$f(x) = \begin{cases} 0, & 0 \leq x^2 < 1 \\ \frac{\ln 2 - \cos 1}{2}, & 9x^2 = 1 \\ -\cos 9x^2, & 9x^2 > 1 \end{cases}$$

discontinuous at $x = \pm \frac{1}{3}$

2. **Ans (1)**

$$y = f(x) = (\ln x)^2 + 2 \ln x$$

$$\left. \frac{dy}{dx} \right|_{A(a, f(a))} = \frac{2 \ln a}{a} + \frac{2}{a} = \frac{2}{a} (1 + \ln a)$$

$$\text{Also } \frac{dy}{dx} = \frac{f(a) - 0}{a - 0}$$

$$\frac{f(a)}{a} = \frac{2}{a} (1 + \ln a) \Rightarrow f(a) = 2(1 + \ln a)$$

$$\text{So } (\ln a)^2 + 2 \ln a = 2 + 2 \ln a$$

$$(\ln a)^2 = 2 \Rightarrow a = e^{\sqrt{2}} \text{ or } e^{-\sqrt{2}}$$

$$\text{Hence } a = e^{-\sqrt{2}} \text{ and } b = e^{\sqrt{2}}$$

$$\text{So } ab = 1$$

3. **Ans (5)**

$$f(x) = 6P - 4P \cos 4x - 5 - 3 \cos 3x > 0$$

$$2P > \frac{5 + 3 \cos 3x}{3 - 2 \cos 4x}$$

$$2P > \text{max value of } \frac{5 + 3 \cos 3x}{3 - 2 \cos 4x}$$

$$2P > \frac{5 + 3}{3 - 2}$$

$$P > 4$$

Smallest P is 5

4. **Ans (8)**

$$g(x) = \frac{2(x^2 - 1)^2}{(x-2)x(2x-1)}$$

$$g'(x) = \frac{4(x^2 - 1)(x^4 - 5x^3 + 6x^2 - 5x + 1)}{(x^2 - 2)x^2(2x - 1)^2}$$

$$g'(x) = 0 \Rightarrow x = \pm 1, 2 \pm \sqrt{3}$$

$$\Rightarrow g(2 + \sqrt{3}) = 8$$

5. **Ans (0)**

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{xf'(x)}{x} = \lim_{x \rightarrow \infty} (f(x) + xf'(x)) = a$$

$$\Rightarrow \lim_{x \rightarrow \infty} xf'(x) = 0$$

$$\text{Now, } \lim_{x \rightarrow \infty} xf'(x) = \frac{x^2 f''(x)}{x};$$

$$\lim_{x \rightarrow \infty} (2xf'(x) + x^2 f''(x)) = 0 \Rightarrow \lim_{x \rightarrow \infty} x^2 f''(x) = 0$$

6. **Ans (9)**

From LMVT, there exists atleast $(n - 1)$ points where $f'(x) = m$, therefore, there exists atleast $(n - 2)$ points where $f''(x) = 0$ (using Rolle's theorem)

7. **Ans (0)**

$$f(x) = 2f\left(\frac{x^2}{2}\right) + (6 - x^2)$$

$$\Rightarrow f'(x) = 2f'\left(\frac{x^2}{2}\right) \cdot x - 2xf'(6 - x^2)$$

$$\Rightarrow f'(x) = 2x \left(f'\left(\frac{x^2}{2}\right) - f'(6 - x^2) \right)$$

$$f'\left(\frac{x^2}{2}\right) > f'(6 - x^2) \text{ (if } \frac{x^2}{2} > 6 - x^2)$$

$\therefore f'(x)$ is increasing

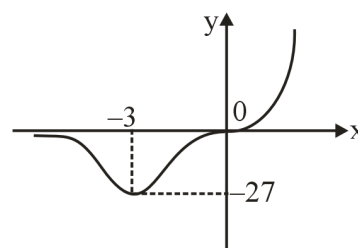
$$\frac{x^2}{2} > 6 - x^2 \Rightarrow x^2 > 4$$

$$\Rightarrow f'\left(\frac{x^2}{2}\right) - f'(6 - x^2) > 0 \text{ when } x < -2 \text{ or } x > 2$$

$$\Rightarrow f'(x) > 0, \text{ when } x \in (-2, 0) \cup (2, \infty)$$

$$\Rightarrow a + b + c = 0$$

8. **Ans (9)**



$$y = x^3 e^{x+3} = -k$$

$$y = x^3 e^{x+3}$$

$$\frac{dy}{dx} = e^{x+3} (3x^2 + x^3) = 0$$

$$\Rightarrow x = 0, x = -3$$

$$\text{at } x = -3, y = -27$$

For 2 distinct real roots, $k \in (0, 27)$

$$2, 3, 5, 7, 11, 13, 17, 19, 23, = 9$$